

Technical Analysis Report

Automated Invoice Data Extraction & Performance Benchmarking

1. Executive Summary

This report evaluates the transition from a vision-based Large Language Model (LLM) approach to a hybrid OCR + LLM pipeline for processing logistics invoices. The primary goal was to eliminate data hallucinations while optimizing operational costs. The project demonstrates that combining Google Document AI for structural text extraction and Gemini 2.0 Flash for semantic reasoning provides the highest level of accuracy for complex, multi-page document classification.

2. Problem Statement

Logistics invoices are notoriously difficult to process due to inconsistent layouts, varying image qualities, and dense technical data (e.g., Forwarder details, Port of Discharge, File Numbers). Initial tests using a pure Vision LLM approach encountered two critical failure points:

- **Character Recognition Sensitivity:** Small fonts or low-DPI scans led to hallucinations in invoice numbers and file references.
- **Entity Confusion:** Models frequently confused "Destination" (the geographical port) with "Recipient" (the legal entity).

3. Comparative Methodology

Method A: Direct LLM Vision (Gemini 2.0 Flash)

This method feeds page images directly to the multimodal model. While efficient for simple layouts, it proved unreliable for precise string matching (e.g., specific ID numbers) across multiple pages.

Method B: Hybrid OCR + LLM Pipeline (Selected Solution)

By decoupling the **Extraction** (Google Document AI) from the **Reasoning** (Gemini 2.0 Flash), we achieved near-perfect reliability. The Document AI processor handles the heavy lifting of OCR, providing a highly accurate text stream that the LLM then parses into structured JSON.

N.B.: A complementary test was conducted using Google's specialized "Invoice Parser" model. While it provided flawless character recognition, occasional entity confusion occurred between "forwarder" and "destination," highlighting the need for an intelligent reasoning layer despite the progress in specialized OCR detection.

4. Detailed Cost Analysis & FinOps Strategy

Approach	OCR Cost (1k p.)	LLM Cost (1k p.)	Total (1k p.)	Performance
Vision-Only (Flash)	\$0.00	~\$0.12	\$0.12	Low Reliability
Specialized Invoice Parser	\$10.00 - \$100.00*	\$0.00	\$10.00+	High Accuracy
Hybrid Pipeline (Selected)	\$1.50	~\$0.35	\$1.85	Optimal / 99%

Cost Insights:

- **The "Rounding Trap":** The specialized Invoice Parser bills in 10-page blocks (\$0.10/block). While a 1000-page document costs \$10, processing 1000 individual 1-page invoices costs \$100.
- **The Hybrid Advantage:** By using Standard OCR (\$1.50/1k pages flat) and Gemini for text analysis, we bypass block-based surcharges, achieving an 81.5% cost reduction compared to the optimal specialized parser rate.

5. Implementation Features

- **Strict Filtering:** Automated rejection of non-invoice pages using icon detection and keyword triggers.
- **Cross-Page Intelligence:** Logic to handle multiple distinct invoices within a single PDF upload.
- **Duplicate Detection:** A hashing algorithm ensuring unique registration of Forwarder/Destination/Invoice Number trios.

6. Conclusion

The Hybrid OCR + LLM approach is the recommended production standard. It delivers superior data integrity through custom-prompted business rules while maintaining a highly competitive cost-to-performance ratio.